# Entity-Aware Generation of Synthetic Clinical Progress Notes for Prostate Cancer using Large Language Models

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## 1 Entity definitions and examples

Thisprovides brief definitions of the entity types used in the entity-level evaluation. Examples are translated into English from detected spans in the annotated clinical texts. For traceability, each example reports the NER model output, case number, and experiment from which the detected span was extracted.

**Age** Mentions of patient age or age-related expressions.

- *Example:* "66 years old".
- *Source:* MedSpaNER xlm-roberta-large-spanish-trials-cases-temp-ent, case 1, E2E few-shot.

**Chemicals** Mentions of pharmacological substances, compounds, biochemical substances, or related biomedical chemicals that can be normalised.

- *Example:* "acetylsalicylic acid".
- *Source:* PharmaCoNER bsc-bio-ehr-es-pharmaconer, case 30, E2E few-shot.

**Date** Calendar dates or relative temporal references that locate a clinical event in time.

- *Example:* "16th postoperative day".
- *Source:* MedSpaNER xlm-roberta-large-spanish-trials-cases-temp-ent, case 63, E2E few-shot.

**Disease** Mentions of diseases, diagnoses, pathological conditions, or clinically established disorders.

- *Example:* "vertebral metastases".

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- *Source:* DisTEMIST bsc-bio-ehr-es-distemist, case 98, E2E few-shot.

**Dose** Quantified medication doses or administered amounts, usually including a numerical value and unit.

- *Example:* "11.25 mg, 2 doses".
- *Source:* MedSpaNER roberta-es-clinical-trials-cases-medic-attr, case 56, E2E few-shot.

**Duration** Expressions indicating the length of time over which a condition, treatment, symptom, or clinical process occurred.

- *Example:* "approximately one year".
- *Source:* MedSpaNER xlm-roberta-large-spanish-trials-cases-temp-ent, case 83, E2E few-shot.

**Frequency** Expressions describing how often an event, symptom, treatment, or clinical action occurs.

- *Example:* "twice per week".
- *Source:* MedSpaNER xlm-roberta-large-spanish-trials-cases-temp-ent, case 1, E2E few-shot.

**Gleason** Mentions of Gleason score or Gleason grading patterns used in prostate cancer assessment.

- *Example:* "Gleason 10 (5+5)".
- *Source:* rule-based Gleason extractor, case 112, E2E few-shot.

**Human** Mentions identifying the patient or another human participant in the clinical narrative.

- *Example:* "male patient".
- *Source:* LivingNER bsc-bio-ehr-es-livingner-humano, case 24, E2E few-shot.

**Neg_cue** Linguistic cues that introduce or signal negation in the clinical text.

- *Example:* in the phrase "without neoplastic involvement", the cue is "without".
- *Source:* MedSpaNER roberta-es-clinical-trials-cases-neg-spec, case 1, E2E few-shot.

**Negated** Clinical entities or findings explicitly stated as absent, excluded, or ruled out within their textual context.

- *Example:* in the phrase "without neoplastic involvement", the negated entity is "neoplastic involvement".
- *Source:* MedSpaNER roberta-es-clinical-trials-cases-neg-spec, case 1, E2E few-shot.

**PSA** Mentions of prostate-specific antigen values, measurements, or references to PSA as a clinical marker.

- *Example:* "244.10 ng/ml".
- *Source:* rule-based PSA extractor, case 24, E2E few-shot.

**Procedure** Mentions of diagnostic, therapeutic, surgical, or radiological procedures.

- *Example:* “adjuvant radiotherapy”.
- *Source:* MedProcNER bsc-bio-ehr-es-medprocner, case 65, E2E few-shot.

**Proteins** Mentions of proteins, biomarkers, or protein-based markers relevant to clinical interpretation.

- *Example:* “androgen receptors”.
- *Source:* PharmaCoNER bsc-bio-ehr-es-pharmaconer, case 82, E2E few-shot.

**Spec_cue** Linguistic cues that introduce uncertainty, suspicion, compatibility, or possibility.

- *Example:* in the phrase “compatible with osteoblastic metastases”, the cue is “compatible with”.
- *Source:* MedSpaNER roberta-es-clinical-trials-cases-neg-spec, case 1, E2E few-shot.

**Speculated** Clinical entities or findings presented as suspected, possible, compatible with, or not fully confirmed.

- *Example:* in the phrase “compatible with osteoblastic metastases”, the speculated entity is “osteoblastic metastases”.
- *Source:* MedSpaNER roberta-es-clinical-trials-cases-neg-spec, case 1, E2E few-shot.

**Species** Mentions of organisms, microorganisms, or biological species relevant to the clinical case.

- *Example:* “Staphylococcus epidermidis”.
- *Source:* LivingNER bsc-bio-ehr-es-livingner-species, case 205, E2E few-shot.

**Symptom** Mentions of symptoms, signs, clinical manifestations, or patient-reported complaints.

- *Example:* “urinary incontinence”.
- *Source:* SympTEMIST bsc-bio-ehr-es-symptemist, case 41, E2E few-shot.

# 2 Prompts

## 2.1 E2E

Listing 1: Prompt used for E2E generation

```
Actúa como un médico especialista en Urología u Oncología que redacta notas evolutivas en una historia clí
    nica electrónica hospitalaria en español.

Se te proporcionará un caso clínico completamente anotado mediante etiquetas XML que identifican entidades
    clínicas (por ejemplo: `<ENFERMEDAD>`, `<Date>`, `<GLEASON>`, etc.). A partir de este caso clínico,
    genera un texto equivalente con el estilo y estructura propios de la narrativa de historia clínica de
    un paciente de cáncer de próstata, tal y como podría aparecer en un sistema de historia clínica real de
     un centro sanitario.

Las posibles etiquetas XML que identifican entidades clínicas son las siguientes: [`<SINTOMA>`, `<
    PROCEDIMIENTO>`, `<ENFERMEDAD>`, `<Age>`, `<Date>`, `<Dose>`, `<Duration>`, `<Frequency>`, `<Neg_cue>`,
     `<Negated>`, `<Spec_cue>`, `<Speculated>`, `<Time>`, `<PROTEINAS>`, `<UNCLEAR>`, `<SPECIES>`, `<HUMAN
    >`, `<NORMALIZABLES>` `<GLEASON>`, `<PSA>`].

El texto generado debe:
```

```
1. Mantener estrictamente la información y los datos clínicamente relevantes desde el punto de vista del manejo médico urológico.
2.Prescindir única y exclusivamente de datos que no tengan relevancia ni significación clínica desde el punto de vista del manejo (diagnóstico, tratamiento, plan de actuación, estadificación, etc.) de la patología de cáncer de próstata.
3. No añadir datos nuevos que no estén explícitamente presentes.
4. No omitir información clínicamente relevante para la patología de cáncer de próstata.
5. No inferir resultados analíticos, radiológicos, de estadiaje, de PSA, de Gleason, ni datos anatomopatológicos si no aparecen en el caso clínico original.
6. Reformular completamente el texto (no copiar frases literalmente) con el estilo y estructura de narrativa de historia clínica.
7. Utilizar estilo clínico habitual en historias clínicas de pacientes reales de cáncer de próstata:
 - Utiliza las frases más breves posibles, directas y clínicamente informativas.
 - Minimiza y condensa al máximo el texto generado, omitiendo todo contenido prescindible, evitando redundancias y priorizando frases breves, compactas y de alta densidad informativa.
 - Evita el lenguaje narrativo o académico del caso clínico original, propio de artículos científicos pero no de historia clínica.
 - No incluyas comentarios explicativos, conectores entre frases ni texto de relleno prescindible.
 - Evita el uso de verbos en las frases siempre que sea posible (p. ej., "Evolución favorable con PSA 0.04 a las 4 semanas" mejor que "Presenta evolución favorable con PSA 0.04 a las 4 semanas").
 - Emplea las abreviaturas clínicas habituales cuando sea apropiado (p. ej., ADC, dx, tto, pte, RT, UCI, ADT, etc.).
 - Utiliza terminología médica habitual en la práctica clínica hospitalaria española.
8. Adapta, en la medida de lo posible, la prosa narrativa extensa del caso clínico original en un formato parecido a este estilo (sin que las secciones se muestre de forma explícita):
 - Información paciente.
 - Diagnóstico inicial + fecha + tratamiento inicial.
 - Progresión + momento temporal + manejo.
 - Complicación/evoluciones + hallazgos + actuación.
 - Situación final + evolución clínica.
9. Mantener coherencia temporal, tal y como está reflejada en el caso clínico original (fechas, intervalos temporales, etc.).
10. Conservar las dosis (sin necesidad específica de mantener las unidades), frecuencias, procedimientos y diagnósticos del caso clínico original.
11. Contener las entidades etiquetadas de la misma manera que se te han proporcionado, es decir siguiendo una estructura de etiquetas similar a XML

No expliques lo que haces. Devuelve únicamente la nota clínica generada.

*Few-shot includes 2 examples here

CASO CLÍNICO ORIGINAL:

{INSERTAR_AQUÍ_EL_TEXTO_DEL_CASO}
```

Listing 2: Prompt used for E2E generation in English

```
Act as a specialist in Urology or Oncology writing clinical progress notes in a hospital Electronic Health Record in English.

You will be provided with a clinical case fully annotated with XML tags identifying clinical entities, such as <DISEASE>, <Date>, <GLEASON>, and related labels. From this annotated clinical case, generate an equivalent text using the style and structure of a clinical progress note for a patient with prostate cancer, as it could appear in a real hospital Electronic Health Record.

The possible XML tags identifying clinical entities are the following: [<SYMPTOM>, <PROCEDURE>, <DISEASE>, <Age>, <Date>, <Dose>, <Duration>, <Frequency>, <Neg_cue>, <Negated>, <Spec_cue>, <Speculated>, <Time>, <PROTEINS>, <UNCLEAR>, <SPECIES>, <HUMAN>, <NORMALIZABLES>, <GLEASON>, <PSA>].

The generated text must:

Strictly preserve all clinically relevant information from the perspective of urological management.
Omit only information that has no clinical relevance for the management of prostate cancer, including diagnosis, treatment, management plan, staging, follow-up or disease status.
Not add any new information that is not explicitly present in the original clinical case.
Not omit information that is clinically relevant to prostate cancer.
Not infer analytical, radiological, staging, PSA, Gleason or pathological findings if they are not present in the original clinical case.
```

```
Fully reformulate the text and avoid copying sentences literally, while preserving the meaning and clinical content.
Use the usual style of real hospital clinical progress notes for prostate cancer patients:
Use the shortest possible sentences, direct and clinically informative.
Condense the generated text as much as possible, avoiding redundancy and prioritising brief, compact and information-dense phrasing.
Avoid the narrative or academic style of the original case report, which is typical of scientific articles but not of clinical records.
Do not include explanatory comments, unnecessary connectors or filler text.
Avoid verbs whenever clinically acceptable, for example: "Favourable evolution with PSA 0.04 at 4 weeks" instead of "The patient presents favourable evolution with PSA 0.04 at 4 weeks".
Use common clinical abbreviations when appropriate, such as ADC, dx, tx, pt, RT, ICU, ADT, and similar terms.
Use standard medical terminology commonly found in hospital clinical documentation.
Adapt the extended narrative prose of the original clinical case into a structure resembling the following format, without explicitly displaying section headings:
Patient information.
Initial diagnosis + date + initial treatment.
Progression + temporal reference + management.
Complications/evolution + findings + clinical action.
Final status + clinical outcome.
Preserve temporal coherence exactly as reflected in the original clinical case, including dates, intervals and temporal sequence.
Preserve doses, frequencies, procedures and diagnoses from the original clinical case.
Preserve the annotated entities using the same XML-like tagging structure provided in the input.

Do not explain your process. Return only the generated clinical note.

Few-shot includes 2 examples here

ORIGINAL CLINICAL CASE:

{INSERT_ORIGINAL_CASE_TEXT_HERE}
```

## 2.2 T2E

Listing 3: Prompt used for T2E generation

```
Actúa como un médico especialista en Urología u Oncología que redacta notas evolutivas en una historia clínica electrónica hospitalaria en español.

Se te proporcionará un caso clínico completamente anotado mediante etiquetas XML que identifican entidades clínicas (por ejemplo: `<ENFERMEDAD>`, `<Date>`, `<GLEASON>`, etc.). A partir de este caso clínico, genera un texto equivalente con el estilo y estructura propios de la narrativa de historia clínica de un paciente de cáncer de próstata, tal y como podría aparecer en un sistema de historia clínica real de un centro sanitario.

Las posibles etiquetas XML que identifican entidades clínicas son las siguientes: [`<SINTOMA>`, `<PROCEDIMIENTO>`, `<ENFERMEDAD>`, `<Age>`, `<Date>`, `<Dose>`, `<Duration>`, `<Frequency>`, `<Neg_cue>`, `<Negated>`, `<Spec_cue>`, `<Speculated>`, `<Time>`, `<PROTEINAS>`, `<UNCLEAR>`, `<SPECIES>`, `<HUMAN>`, `<NORMALIZABLES>` `<GLEASON>`, `<PSA>`].

El texto generado debe:

1. Mantener estrictamente la información y los datos clínicamente relevantes desde el punto de vista del manejo médico urológico.
2.Prescindir única y exclusivamente de datos que no tengan relevancia ni significación clínica desde el punto de vista del manejo (diagnóstico, tratamiento, plan de actuación, estadificación, etc.) de la patología de cáncer de próstata.
3. No añadir datos nuevos que no estén explícitamente presentes.
4. No omitir información clínicamente relevante para la patología de cáncer de próstata.
5. No inferir resultados analíticos, radiológicos, de estadiaje, de PSA, de Gleason, ni datos anatomopatológicos si no aparecen en el caso clínico original.
6. Reformular completamente el texto (no copiar frases literalmente) con el estilo y estructura de narrativa de historia clínica.
7. Utilizar estilo clínico habitual en historias clínicas de pacientes reales de cáncer de próstata:
```

```
 - Utiliza las frases más breves posibles, directas y clínicamente informativas.
 - Minimiza y condensa al máximo el texto generado, omitiendo todo contenido prescindible, evitando
     redundancias y priorizando frases breves, compactas y de alta densidad informativa.
 - Evita el lenguaje narrativo o académico del caso clínico original, propio de artículos científicos pero
     no de historia clínica.
 - No incluyas comentarios explicativos, conectores entre frases ni texto de relleno prescindible.
 - Evita el uso de verbos en las frases siempre que sea posible (p. ej., "Evolución favorable con PSA 0.04 a
      las 4 semanas" mejor que "Presenta evolución favorable con PSA 0.04 a las 4 semanas").
 - Emplea las abreviaturas clínicas habituales cuando sea apropiado (p. ej., ADC, dx, tto, pte, RT, UCI, ADT
     , etc.).
 - Utiliza terminología médica habitual en la práctica clínica hospitalaria española.
8. Adapta, en la medida de lo posible, la prosa narrativa extensa del caso clínico original en un formato
     parecido a este estilo (sin que las secciones se muestre de forma explícita):
 - Información paciente.
 - Diagnóstico inicial + fecha + tratamiento inicial.
 - Progresión + momento temporal + manejo.
 - Complicación/evoluciones + hallazgos + actuación.
 - Situación final + evolución clínica.
9. Mantener coherencia temporal, tal y como está reflejada en el caso clínico original (fechas, intervalos
     temporales, etc.).
10. Conservar las dosis (sin necesidad específica de mantener las unidades), frecuencias, procedimientos y
     diagnósticos del caso clínico original.
11. Contener las entidades etiquetadas de la misma manera que se te han proporcionado, es decir siguiendo
     una estructura de etiquetas similar a XML

No expliques lo que haces. Devuelve únicamente la nota clínica generada.

*Few-shot includes 2 examples here

CASO CLÍNICO ORIGINAL:

{INSERTAR_AQUÍ_EL_TEXTO_DEL_CASO}
```

Listing 4: Prompt used for T2E generation in English

```
Act as a specialist in Urology or Oncology writing clinical progress notes in a hospital Electronic Health
     Record in English.

You will be provided with a prostate cancer clinical case written as a biomedical case report. From this
     clinical case, generate an equivalent text using the style and structure of a clinical progress note
     for a patient with prostate cancer, as it could appear in a real hospital Electronic Health Record.
     During generation, you must also identify and annotate the clinically relevant entities using XML-like
     tags.

The possible XML tags identifying clinical entities are the following: [<SYMPTOM>, <PROCEDURE>, <DISEASE>, <
     Age>, <Date>, <Dose>, <Duration>, <Frequency>, <Neg_cue>, <Negated>, <Spec_cue>, <Speculated>, <Time>,
     <PROTEINS>, <UNCLEAR>, <SPECIES>, <HUMAN>, <NORMALIZABLES>, <GLEASON>, <PSA>].

The generated text must:

Strictly preserve all clinically relevant information from the perspective of urological management.
Omit only information that has no clinical relevance for the management of prostate cancer, including
     diagnosis, treatment, management plan, staging, follow-up or disease status.
Not add any new information that is not explicitly present in the original clinical case.
Not omit information that is clinically relevant to prostate cancer.
Not infer analytical, radiological, staging, PSA, Gleason or pathological findings if they are not present
     in the original clinical case.
Fully reformulate the text and avoid copying sentences literally, while preserving the meaning and clinical
     content.
Use the usual style of real hospital clinical progress notes for prostate cancer patients:
Use the shortest possible sentences, direct and clinically informative.
Condense the generated text as much as possible, avoiding redundancy and prioritising brief, compact and
     information-dense phrasing.
Avoid the narrative or academic style of the original case report, which is typical of scientific articles
     but not of clinical records.
Do not include explanatory comments, unnecessary connectors or filler text.
Avoid verbs whenever clinically acceptable, for example: "Favourable evolution with PSA 0.04 at 4 weeks"
     instead of "The patient presents favourable evolution with PSA 0.04 at 4 weeks".
```

```
Use common clinical abbreviations when appropriate, such as ADC, dx, tx, pt, RT, ICU, ADT, and similar terms
     .
Use standard medical terminology commonly found in hospital clinical documentation.
Adapt the extended narrative prose of the original clinical case into a structure resembling the following
     format, without explicitly displaying section headings:
Patient information.
Initial diagnosis + date + initial treatment.
Progression + temporal reference + management.
Complications/evolution + findings + clinical action.
Final status + clinical outcome.
Preserve temporal coherence exactly as reflected in the original clinical case, including dates, intervals
     and temporal sequence.
Preserve doses, frequencies, procedures and diagnoses from the original clinical case.
Annotate the clinically relevant entities in the generated note using the XML-like tag set defined above.
     The tags must be applied directly to the corresponding entity spans in the generated text.

Do not explain your process. Return only the generated clinical note.

Few-shot includes 2 examples here

ORIGINAL CLINICAL CASE:

{INSERT_ORIGINAL_CASE_TEXT_HERE}
```

# 3 Original source case for E2E & T2E

| Original clinical case with RAG | Original clinical case without RAG |
|---|---|
| <HUMAN>Male</HUMAN> aged <Age>82 years</Age> with a history of <PROCEDURE>transurethral resection of the prostate</PROCEDURE> <Date>10 years ago</Date>. The <HUMAN>patient</HUMAN> presents with <DISEASE>recurrent urinary tract infections</DISEASE> and severe <SYMPTOM>lower urinary tract symptoms</SYMPTOM> since <Date>several months ago</Date>. Due to the poor clinical course, <PROCEDURE>bladder ultrasound</PROCEDURE> is performed, showing a large <SYMPTOM>solid mass</SYMPTOM> with an estimated volume of 400 cc, composed of solid stroma with some areas of calcification and <DISEASE>hematoma</DISEASE>. <Speculation_cue>It appears</Speculation_cue> to show <Speculated><DISEASE>dependence on prostatic tissue</DISEASE></Speculated>. <PROCEDURE>computed tomography</PROCEDURE> (<PROCEDURE>CT</PROCEDURE>) is performed, confirming the presence of a large <SYMPTOM>pelvic mass arising from the prostate</SYMPTOM> with bladder involvement. <PROCEDURE>excision of the <SYMPTOM>mass</SYMPTOM></PROCEDURE> is performed, with a <PROCEDURE>pathological diagnosis</PROCEDURE> of <DISEASE>prostatic cyst</DISEASE>. | An 82-year-old male with a history of transurethral resection of the prostate 10 years ago. The patient presents with recurrent urinary tract infections and severe lower urinary tract symptoms for several months. Due to the poor clinical course, bladder ultrasound is performed, showing a large solid mass with an estimated volume of 400 cc, composed of solid stroma with some areas of calcification and hematoma. It appears to show dependence on prostatic tissue. Computed tomography (CT) is performed, confirming the presence of a large pelvic mass arising from the prostate with bladder involvement. Excision of the mass is performed, with a pathological diagnosis of prostatic cyst. |

Figure C1: Qualitative example for patient 2 illustrating the main observable difference between E2E and T2E. The left column shows the original clinical case with XML-style entity annotations used by the RAG-based E2E configuration, whereas the right column shows the original clinical case for T2E with the same clinical content expressed as plain text.

# 4 LLMs, Models, Cost and Usage

- **GPT-5.4 Nano (OpenAI)** was included as a proprietary cloud-based model accessed through OpenAI API. Since it is a closed-source system, architectural details and parameter count are not publicly disclosed. It was selected because of its efficient inference profile, large-context capabilities and suitability for instruction-following generation tasks requiring controlled clinical reformulation. Total cost for generating 436 evolution notes was 1.14$ and a total of 1.35 Million tokens (input & output).

- **Qwen 3.5:35B A3B** was used as an open-weights model for local inference. Its A3B configuration provides an optimised deployment setting for efficient generation while preserving strong multilingual and reasoning capabilities. Local execution offers greater control over inference parameters and reduces exposure of input data to external services. In this study, the model was deployed on two dedicated GPU instances (vía Ollama) with 32GB of VRAM, achieving a mean latency of $\approx$ 7 seconds per patient and a throughput of $\approx$ 75 tokens per second. Its instruction-following capacity makes it suitable for constrained clinical generation, where the output must preserve source information, relevant entities and temporal coherence.

- **GLM-5** was selected as a large multilingual general-purpose model based on a Mixture-of-Experts architecture. This design enables dynamic routing through specialised expert components, improving scalability for complex generation tasks. Given its large parameter scale, inference was performed through cloud access. The model was included to assess the behaviour of a high-capacity multilingual system in the generation of clinically constrained prostate cancer evolution notes. Model was deployed on Ollama using payed *Pro Plan*. Exact number of used token is undisclosed.

- **Claude Sonnet 4.6 (CS4.6 - Anthropic)** was included as a proprietary cloud-based generation model accessed through Anthropic API. It was selected to complement the comparison with another closed-source system with strong instruction-following and long-context capabilities. In this study, Claude Sonnet 4.6 was used to generate synthetic clinical evolution notes under the same task constraints as the other generative models, allowing comparison across proprietary, open-weights, local and cloud-based systems. Total cost for generating 436 evolution notes was 11.65$ and total usage of 1.737.783 input tokens and 429.244 output tokens.

- **BioMistral 7B** was considered as a biomedical domain-adapted baseline [**?**]. Its inclusion was intended to examine whether biomedical pre-training provided advantages over larger general-purpose models in this constrained generation task. However, its outputs did not reach the minimum quality threshold established for the study, particularly in terms of clinical fidelity and structural adequacy. For this reason, BioMistral 7B was excluded from the main comparative analysis.

- **Kimi K2.6** was used as the independent LLM-as-a-judge evaluator accessed via MoonShot API. Its role was not to generate clinical notes, but to provide structured assessments of the generated outputs according to the evaluation criteria defined in the study. Using a separate evaluator model ensured that the quality assessment was not performed by any of the models involved in generation, thereby reducing self-evaluation bias. The cost to evaluate all generated notes (1.744) has been 4.89586$.

  The total cloud models cost throughout this research was 37.68$.

# 5 Used Metrics Formula

Precision, recall and F1-score were then computed as:

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \quad F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}.$$

Since true negatives are not defined in span-based NER evaluation, accuracy was calculated as an entity-level agreement score over the union of matched and unmatched annotations:

$$Accuracy = \frac{TP}{TP + FP + FN}.$$

These metrics were used to quantify entity-level agreement between projected and silver-standard annotations in the generated notes. The resulting scores enabled comparison between the RAG-based setting (E2E) and the non-RAG setting (T2E), with particular emphasis on entity preservation, localisation and recoverability.

# 6 Error Exemplification

| **Original clinical case** | **Generated synthetic clinical progress note** |
|---|---|
| A 79-year-old man, with performance status 0, presented with a PSA level of 11 ng/mL. A transrectal biopsy was performed, showing Gleason 4+5=9 adenocarcinoma with signet-ring cell features. Laparoscopic radical prostatectomy with bilateral pelvic lymphadenectomy was performed. Pathological analysis showed Gleason 4+5=9 prostate adenocarcinoma, pT2cN0M0, with signet-ring cell features. The tumor expressed for PSAP, PSA, and PAS, and was negative for CEA. One month after surgery, the PSA level was 4.8 ng/mL, and treatment with goserelin plus bicalutamide was initiated for two years. The patient was lost to follow-up at 37 months, with a last recorded PSA level of 0.32 ng/mL. | A 79-year-old man, PS 0, presented due to elevated PSA levels of 11 ng/mL.<br>Transrectal biopsy showed Gleason 4+5=9 prostate adenocarcinoma with signet-ring cell differentiation. Staging was pT2cN0M0.<br>Surgical treatment consisted of laparoscopic radical prostatectomy with bilateral pelvic lymphadenectomy. Pathological examination showed Gleason 4+5=9 prostate adenocarcinoma with signet-ring cell morphology, pT2cN0M0. Immunohistochemistry showed positivity for PSAP and negativity for CEA, PSA, and PAS.<br>One month after surgery, the PSA level was 4.8 ng/mL. Given the absence of an adequate nadir, complete androgen blockade with goserelin plus bicalutamide was initiated for two years.<br>At 37 months, the last recorded PSA level was 0.32 ng/mL. The patient was lost to follow-up at that time. |

Figure D1: Qualitative example for patient 113 illustrating the main observable safety error. In red, marked error regarding original clinical case report. Generated synthetical clinical progress note corresponds to CS 4.6 on E2E zero-shot configuration. Other 5 experiments—from CS4.6-E2E-few-shot, GLM5-E2E-few, GPT5.4-nano-E2E-zero-shot, Qwen3.5:35b-E2E-few-shot and Qwen3.5:35b-E2E-zero-shot prompting configurations—exhibited the exact same error.